

Hardness, Toughness, and Scratchability of Germanium–Selenium Chalcogenide Glasses

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Ge–Se chalcogenide glasses are characterized by relatively low hardness (0.39–2.35 GPa) and low fracture toughness (0.1–0.28 MPa·m^{1/2}). Actually, the hardness of chalcogen-rich glasses is low enough so that the brittleness parameter, $B = H/K_c$, is lower than that of silicate glasses. Whereas hardness and Young's modulus increase with increasing germanium contents, fracture toughness follows a trend similar to that of the density and exhibits a maximum for the Ge₂₀Se₈₀ composition, which corresponds to the rigidity percolation threshold. Optical microscopy and atomic force microscopy observations suggest that the indentation deformation proceeds by a localized shear deformation phenomenon. Glasses in the chalcogen-rich region behave viscoelastically at room temperature. As a consequence, an increase of the loading time results in a decrease of hardness and toughness.

I. Introduction

INTEREST in chalcogenide glasses lies primarily in their transparency in the second atmospheric window, from 8 to 14 μm.^{1,2} Various optical devices, using lens and fibers, have been successfully produced for application in night visibility and in thermometry. However, these glasses suffer from a high brittleness (low fracture toughness), which makes the fabrication and the manipulation of structural parts very delicate and decreases the life duration in service. Therefore, it is of paramount importance to study the resistance of chalcogenide glasses against surface damage due to sharp mechanical contacts. From a more academic viewpoint, chalcogenides glasses are interesting because of (i) their strong covalent character of interatomic bonding and (ii) their relatively low softening temperatures, which offer the possibility to study viscoelasticity as well as relaxation processes at or a little above room temperature.

II. Materials and Experimental Methods

Glasses were synthesized by introducing the appropriate quantities of germanium (Ge) and selenium (Se) in silica (SiO₂) ampoules sealed under vacuum, by further heating this mixture at high temperature, typically between 750° and 950°C, and by

subsequent quenching (see Ref. 3 for details). The glass-transition temperatures, T_g , were determined using differential thermal analysis (DTA) at a heating rate of 10°C/min.

The elastic moduli were calculated from the measurements of the longitudinal, V_l , and transverse, V_t , ultrasonic wave velocities with a relative error of better than ±2% using 10 MHz piezoelectric transducers. The Young's modulus, E , bulk modulus, K , and Poisson's ratio, ν , were derived from the classical elasticity relationships:

$$E = \rho \frac{3V_l^2 - 4V_t^2}{(V_l/V_t)^2 - 1} \quad (1)$$

$$\nu = \frac{3V_l^2 - 4V_t^2}{2(V_l^2 - V_t^2)} - 1 \quad (2)$$

$$K = \frac{E}{3(1 - 2\nu)} \quad (3)$$

where ρ is the density of the material.

The density was measured at 20°C using the Archimedes displacement technique with CCl₄. The relative error on the density was ±0.5%. The indentation behavior was investigated using a diamond Vickers indenter. Loads between 0.098 and 49.05 N were applied for 20 s. All the characteristics were averaged over measurements on 15 indentations per grade. Specimen surfaces were mirror polished with alumina suspensions up to 0.25 μm particle size before indentation. All the measurements were performed in a thermally regulated room, at 20°C. This latter condition was of primary importance because selenium-rich glasses have T_g as low as 41°C and exhibit a temperature-dependent behavior.⁴ Vickers (H_v) and Meyer's (H) hardness are defined by

$$H_v = \frac{0.4636P}{a^2} \quad (4)$$

$$H = \frac{P}{2a^2} \quad (5)$$

where P (in newtons) is the load applied on the indenter and a (in meters) is the half of the mean size of the two diagonals. The experimental error on H , resulting from the inaccuracy on the measurement of a , was about ±0.03 GPa. H , analogous to a mean normal stress, was the more fundamental measure of hardness. The apparent indentation fracture toughness, K_c , of the glass was determined from the mean size of the cracks observed on the surface of the specimen growing away from the indentation corners. We let c be half the distance between two opposite crack tips; then, the load was carefully chosen to ensure $c/a > 2.5$, so that a radial-median crack pattern prevailed and c was the radius of

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the half-penny-like cracks (Fig. 1). The following equation was used:⁵

$$K_c = 0.016 \left(\frac{E}{H} \right)^{1/2} \frac{P}{c^{3/2}} \quad (6)$$

The topology of the indented surfaces was analyzed using either optical microscopy or atomic force microscopy (AFM; Model Nanoscope III, Digital Instruments, Tonawanda, NY) operated in tapping and contact modes, depending on the scale of the observation. The scatter on the fracture toughness data was mainly due to the statistical nature of fracture, which especially affected the results because chalcogenide glasses are remarkably brittle.

The scratch resistance was characterized using laboratory-designed instrumented scratch-testing equipment.⁶ The mirror-polished specimen was pasted on a flat horizontal support and translated at a constant speed of 10 $\mu\text{m/s}$. The load was applied vertically using a conic (apex angle = 90°) diamond indenter, at a constant loading rate of 0.01 N/s, from 0 to 2 N, over a scratch distance of 2 mm. The normal and the tangential (or scratch resistance) forces were continuously monitored and each measurement was done three times to investigate the reproducibility. Optical microscopy allowing for interferometrical observations was used to analyze the surface damage. Results obtained on three compositions of Ge–Se glass, i.e., $\text{Ge}_{10}\text{Se}_{90}$, $\text{Ge}_{20}\text{Se}_{80}$, and $\text{Ge}_{30}\text{Se}_{70}$, were very reproducible.

III. Composition Dependence of the Mechanical Properties

(1) Preliminary Considerations on Ge–Se Glass Structure

In what follows, the properties discussed are based on available information pertaining to the atomic or molecular structure of Ge–Se glass networks. Therefore, it is necessary to introduce some basic considerations. The structure of chalcogenide glasses has much in common with the structure of polymers. The structure consists of chalcogen chains, $[-\text{Se}]_n$, that are cross-linked by fourfold (germanium) metalloid atoms. In such glasses, the amount of cross-linking is important. The average coordination number, defined by⁷

$$\langle r \rangle = 2(x + 1) \quad (7)$$

allows for a simple description of the expected amount of cross-linking of a $\text{Ge}_x\text{Se}_{1-x}$ glass network ($x \leq 1$), provided germanium and selenium are fourfold and twofold coordinated,

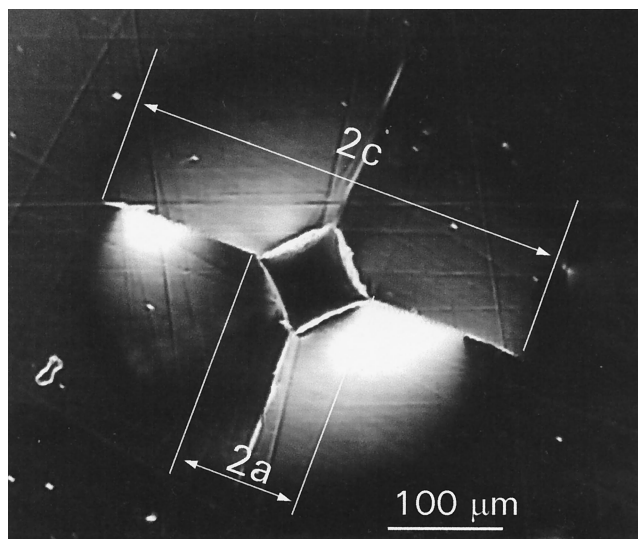


Fig. 1. Typical indentation cracking pattern and characteristic lengths a and c used in the calculation of H and K_c .

respectively. Following this approach, transitions in physical properties should be observed at given $\langle r \rangle$ values. For $\langle r \rangle$ values of < 2.1 , the fraction of chains containing germanium is negligible, and deformation can occur through the alignment of selenium chains with principal tensile stresses or normal to principal compressive stresses. In this range, the deformation resistance is assumed to originate from interchain van der Waals forces, as long as the chains do not interweave. For $\langle r \rangle$ values between 2.1 and 2.2, germanium atoms form bridges between selenium chains and result in a two-dimensional structure. For $\langle r \rangle$ values between 2.2 and 2.4 (corresponding to the GeSe_4 stoichiometry), the glass network is built on GeSe_4 units, two germanium atoms being separated by selenium chains of at least two atoms. It has been suggested that this structure organizes to form trilayer units, where the two outside layers contain the chalcogen anions and the central layer contains germanium. A so-called rigidity percolation threshold is associated with $\langle r \rangle = 2.4$, which corresponds to a fully cross-linked network with two selenium atom segments between germanium atoms. With increasing $\langle r \rangle$ values, the two-dimensional layer structures spread and are fully evolved at $\langle r \rangle = 2.67$ (GeSe_2 stoichiometry), where a so-called chemical threshold occurs as Ge–Se bonds begin to form, starting a transition to three-dimensional network structures.^{7,8} Figure 2 shows schematic drawings of the typical molecular arrangements in Ge–Se glasses for various values of $\langle r \rangle$. Selenium atoms leave lone-pair electrons, which results in weak intermolecular forces that govern the mechanical properties in the chalcogen-rich region. As $\langle r \rangle$ increases, these forces are gradually dominated by stronger covalent bonds (intramolecular Se–Se, Se–Ge, and Ge–Ge bonds). Strong supports to the former structural considerations have been recently provided by neutron diffraction studies and extended X-ray absorption fine structure (EXAFS) results.^{9,10} In addition to the former simple considerations, the glass network structure is complicated by the possible formation of Se_8 rings¹¹ in the chalcogen-rich region and by smaller ring structures for $\langle r \rangle$ between 2.4 and 2.67, because layer structures are favorable in creating small rings.⁸ These small rings may lead to a nanoheterogeneous glass network (nanophase separation) and may be effective in increasing the rigidity percolation threshold from $\langle r \rangle = 2.4$,

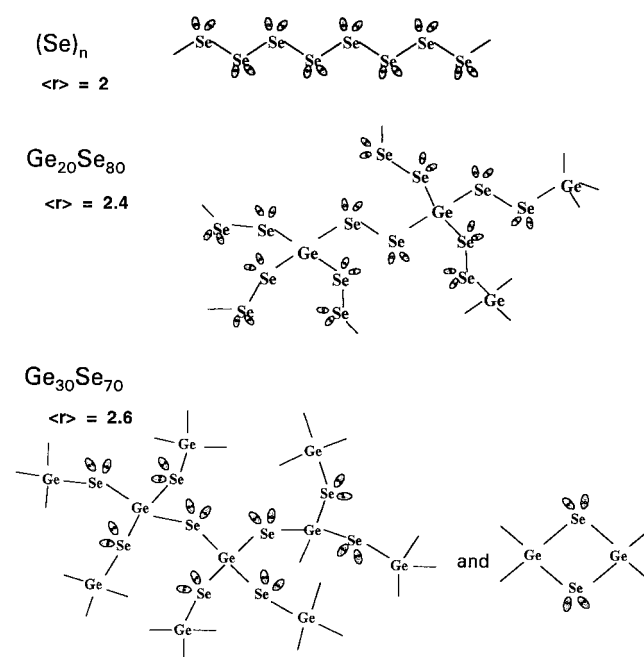


Fig. 2. Schematic drawings of the typical structure of Ge–Se glasses at the atomic scale. Note the lone-pair electrons of the selenium atoms.

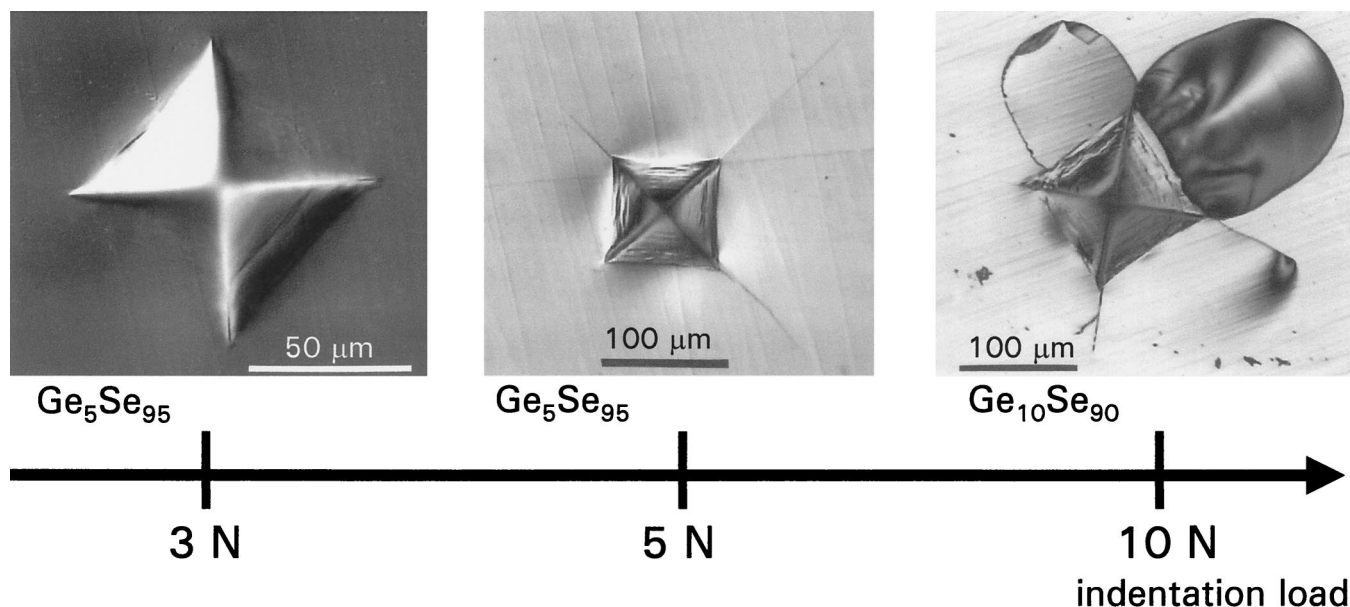


Fig. 3. Changes observed in the indentation cracking pattern of Ge–Se glasses with increasing load (optical microscopy).

corresponding to the composition threshold, to $\langle r \rangle = 2.6$, in agreement with experimental observation.¹²

(2) Indentation Fracture Behavior

Ge–Se glasses behave normally with respect to indentation; i.e., radial–median cracks tend to form under Vickers indentation. For a given Ge–Se glass composition, the indentation cracking behavior depends much on the applied load. The various crack patterns observed are illustrated in Fig. 3. There are typically three major microcracking regimes: (i) below a critical indentation load, hereafter noted P^* , no crack is observed; (ii) above P^* , two cracks appear, symmetrical with respect to the indentation, which correspond to the intersect of a half-penny shape crack with the surface, as has been demonstrated by postindentation polishing of the surface (as the load increases, the number of cracks connected to the indentation corners increases to three and four); and (iii) above a critical load, hereafter noted P_c , chipping occurs. P_c usually almost coincides with the load necessary for the four-crack pattern to be observed. A detailed analysis of the indentation corners reveals the nucleation of several small cracks, typically $<10 \mu\text{m}$ in length, for indentation loads at or just below P^* (Fig. 4). AFM investigations of the surface of the inverse pyramids show that the observed straight lines are not open cracks but shear-induced steps, parallel to the indentation edges (Fig. 5). These steps, $\sim 50 \text{ nm}$ high, reflect a localized shear deformation process. In other words, Ge–Se glasses behave in a very normal way, with little or no flow-compaction contribution, and it seems that the higher the germanium content, the easier the shear lines form. The load ranges for P^* and P_c are summarized in Table I and are plotted as a function of the germanium content in Fig. 6. The sensitivity of the glass to crack formation or to chipping increases significantly with germanium content. Lawn and Marshall¹³ and Hagan¹⁴ have suggested that the critical load for crack initiation from a preexisting subsurface flaw is proportional to K_c^4/H^3 . According to our experimental data, the proportionality factor is $\sim 500 \times 10^3$, whereas coefficients of 22×10^3 and 885 have been proposed by Lawn and Hagan, respectively, from data obtained on either oxide glasses or ceramic materials. Furthermore, the former dependence of P^* on K_c and H predicts a linear increase of P^* with the germanium content up to 20%, whereas the opposite trend is observed. This result requires further analysis of the flaw

characteristics and of the possible incidence of the glass network structure.

(3) Indentation Hardness

Hardness measurements have been found independent of the applied load. The hardness of Ge–Se glasses, as calculated from Eqs. (4) and (5), is given as a function of the composition in Table II. Ge–Se glasses are characterized by relatively low hardness values in comparison with most other glasses. Starting from a value as low as 0.39 GPa for a pure selenium glass ($\sim 5.5 \text{ GPa}$ for a standard soda–lime–silica glass), hardness increases monotonically with increasing germanium contents,

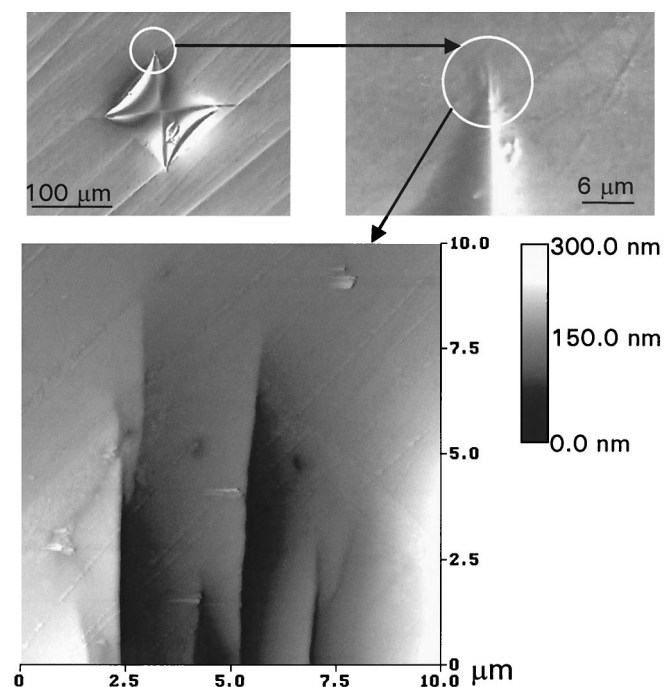


Fig. 4. Optical micrograph showing the indentation pattern obtained at P^* in the $\text{Ge}_{10}\text{Se}_{30}$ glass, with an enlargement of the nucleated microcracks at a corner of the indent (AFM, tapping mode).

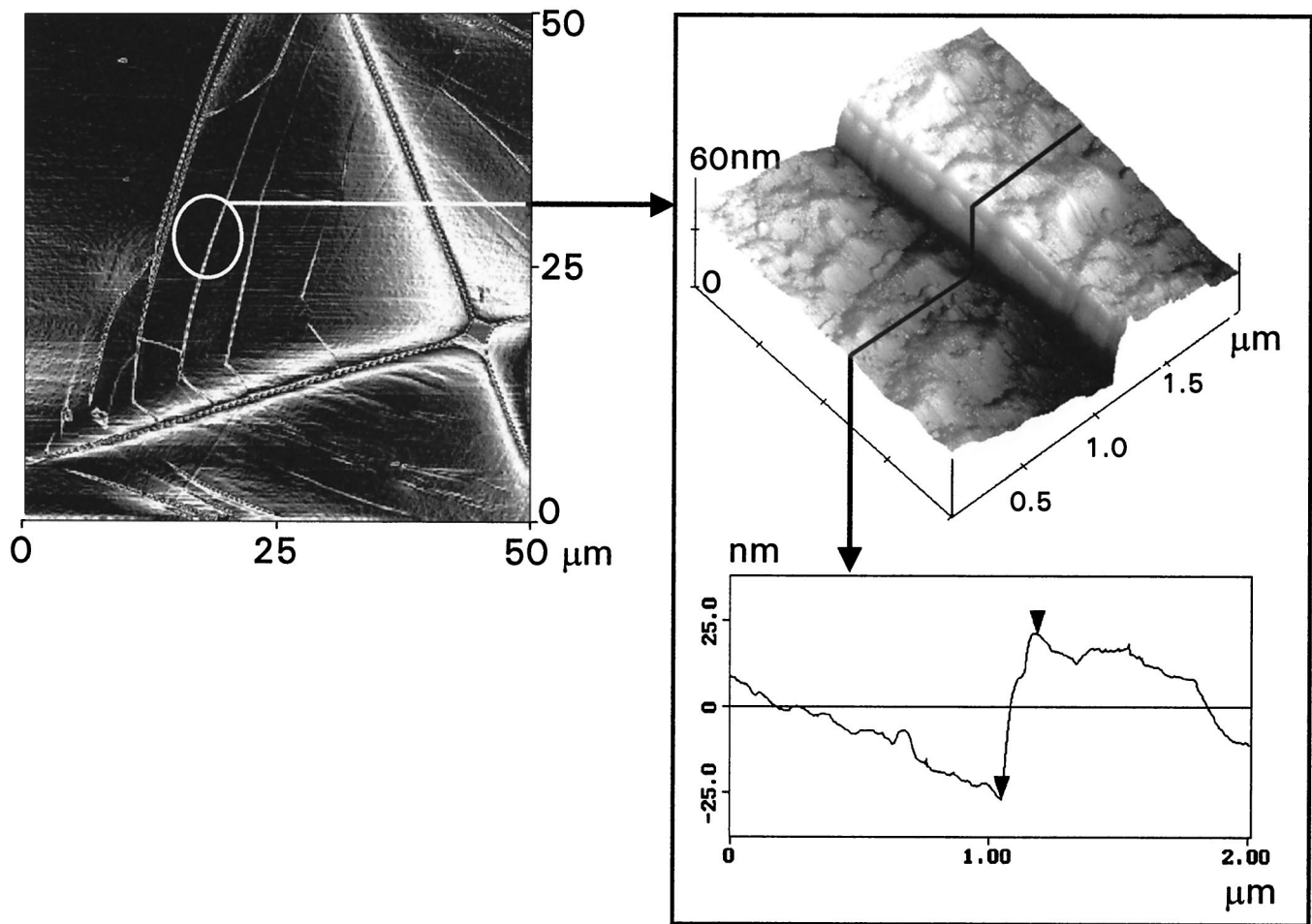


Fig. 5. Shear-deformation-induced steps observed on the indentation surface of a $\text{Ge}_{15}\text{Se}_{85}$ specimen indented with a 4.9 N load for 20 s. Insert shows a typical step profile (AFM, contact mode).

as previously reported by Shkol'nikov,¹⁵ Giridhar *et al.*,¹⁶ and Sreeram *et al.*¹⁷ Hardness is plotted as a function of $\langle r \rangle$ in Fig. 7. In agreement with Sreeram *et al.*, we come to the conclusion that H changes smoothly with $\langle r \rangle$, and no transition is observed. The hardness curve can be readily extrapolated beyond the glassforming region (i.e., for $\langle r \rangle \leq 2.8$) to meet the data corresponding to a germanium polycrystal. Although our data are consistent with those obtained by Sreeram *et al.*, we cannot fit the data with a straight line. Because the Se–Se and Se–Ge bond strengths are similar (223 and 230 kJ/mol, respectively) and the Ge–Ge bond strength is slightly weaker (186 kJ/mol),¹⁵ the source for the increase of the hardness is supposed to be topological in nature. Thus, the gradual increase of the hardness is of the same origin as the increase of the elastic moduli, and it results from the increasing number of strong covalent bonds with increasing $\langle r \rangle$. The indentation hardness depends on loading duration (Fig. 8). This phenomenon originates from the low refractoriness of Ge–Se glasses ($T_g = 41^\circ\text{--}340^\circ\text{C}$). Glasses in the chalcogen-rich region behave viscoelastically at room

temperature, as has been observed by Lewis *et al.*¹⁸ A direct consequence is that the size of an indentation increases with time, i.e., hardness decreases with increases in loading time. The indentation creep phenomenon observed in Ge–Se glasses results from the low activation energy for flow and the significant viscosity decrease beneath the indenter because of a shear-thinning behavior. This complicated topic has been addressed in detail in a companion paper.¹⁹

(4) Indentation Fracture Toughness

K_c can be estimated from Eq. (6). Values for Young's modulus and K_c are summarized in Table II. Ge–Se glasses are characterized by relatively low fracture toughness values in comparison to most other glasses (typical values for soda–lime–silica glasses are $\sim 0.75 \text{ MPa}\cdot\text{m}^{1/2}$). No static fatigue phenomenon is observed, probably because neither germanium nor selenium is soluble in water or reacts with water vapor at room temperature. The dependence of K_c on the germanium content

Table I. Load Ranges where Transition in the Indentation Fracture Behavior are Observed

	Load range (N) for various Ge/Se [†]						
	0/100	5/95	10/90	15/85	20/80	25/75	30/70
Onset of radial crack formation, $P^{*†}$	4.9–9.8	0.98–1.96	0.42–0.98	0.25–0.49	ND	0.25–0.49	0.1–0.25
Onset of chipping, P_c	ND	9.8	4.9	1.96–4.9 [§]	ND	2.94	0.98

[†]ND is not determined. [‡]Five experiments for each load. [§]Chips were observed to form from the edges of the indentation above 1.96 N.

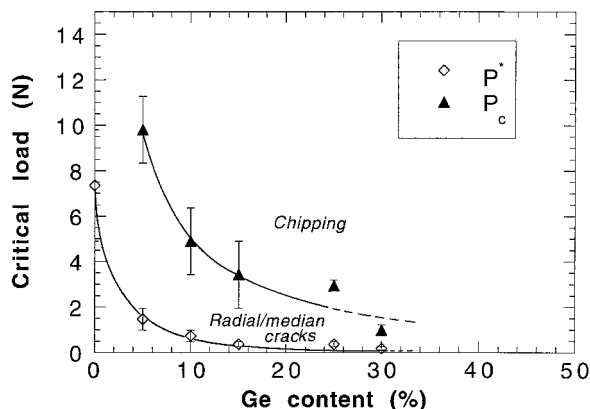


Fig. 6. Load ranges for the various microcracking regimes as a function of the germanium content.

exhibits a maximum for the GeSe_4 composition (Fig. 9), corresponding to $\langle r \rangle = 2.4$ and to the rigidity percolation threshold beyond which Ge–Se–Ge bridges form. Figure 9 shows that pure selenium results in a partially devitrified material so that the K_c value for this grade should be disregarded. Whereas changes in hardness reflect global changes in the material behavior, following changes in the elastic moduli for instance, fracture toughness is related to the fine details of crack nucleation and propagation. Therefore, it can be anticipated that increases or decreases in fracture toughness are due to either (i) crack-tip blunting phenomena caused by, for instance, viscoelastic stress relaxation, or (ii) progressive changes of the crack path in the glass network, associated with changes of the energy required per unit of crack surface area. Hardness testing provides evidence for a viscoelastic behavior of Ge–Se glass at room temperature (see previous paragraph), especially for selenium-rich glasses. This phenomenon is currently under survey. A tentative analysis of the fracture toughness change that is expected from a theoretical estimation of the energy required to form new surfaces during the fracture process is now presented. Let us consider the $\text{Ge}_x\text{Se}_{1-x}$ composition. The number of such units per unit of volume is $\rho N/M$, where ρ is the density, N the Avogadro number, and M the molar mass ($M = 72.6x + 79(1 - x)$). Therefore, the number of $\text{Ge}_x\text{Se}_{1-x}$ units per unit area is given by $(\rho N/M)^{2/3}$. As long as Ge–Ge bonds do not form, i.e., presumably for $x \leq 1/3$, the numbers of Ge–Se and Se–Se bonds expected to form for a $\text{Ge}_x\text{Se}_{1-x}$ unit are $4x$ and $1 - 3x$, respectively, assuming germanium and selenium atoms are fourfold and twofold coordinated, respectively. However, these numbers of bonds again must be understood to be per volume of the $\text{Ge}_x\text{Se}_{1-x}$ unit, and, therefore, should be decreased to the power $2/3$ to get

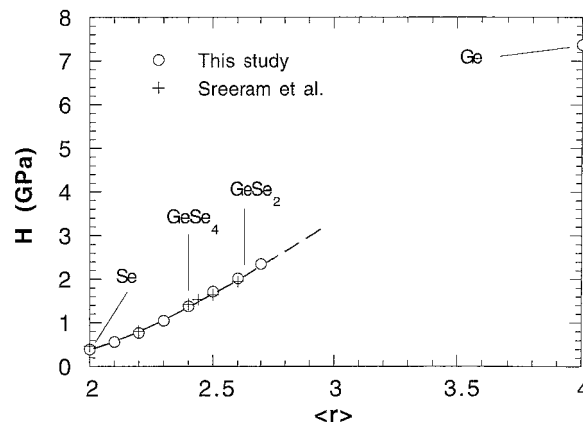


Fig. 7. Hardness (Vickers indentation) of Ge–Se glasses as a function of the average coordination number, $\langle r \rangle$.

a surface equivalent. Therefore, the surface energy required to create new random surfaces through a bond-breaking process is expressed as

$$\gamma_{\text{theo}} \left(x \leq \frac{1}{3} \right) = \frac{1}{2} \frac{1}{N} [(4x)^{2/3} U_{\text{Ge-Se}} + (1 - 3x)^{2/3} U_{\text{Se-Se}}] \left(\frac{\rho N}{M} \right)^{2/3} \quad (8)$$

The $1/2$ prefactor in the right-hand term is introduced to account for the fact that the a crack has two lips.

For $x \geq 1/3$, a similar analysis shows that there are $2(1 - x)$ and $(3x - 1)$ Ge–Se and Ge–Ge bonds, respectively, for a $\text{Ge}_x\text{Se}_{1-x}$ unit, so that the theoretical fracture surface energy is given by

$$\gamma_{\text{theo}} \left(x \geq \frac{1}{3} \right) = \frac{1}{2} \frac{1}{N} [(2 - 2x)^{2/3} U_{\text{Ge-Se}} + (3x - 1)^{2/3} U_{\text{Ge-Ge}}] \left(\frac{\rho N}{M} \right)^{2/3} \quad (9)$$

where, in Eqs. (8) and (9), the used interatomic bond energies are taken from the literature: ($U_{\text{Se-Se}}$, $U_{\text{Ge-Se}}$, $U_{\text{Ge-Ge}}$) = (223, 230, 186) kJ/mol from Ref. 15 or ($U_{\text{Se-Se}}$, $U_{\text{Ge-Se}}$, $U_{\text{Ge-Ge}}$) = (184, 215, 157) kJ/mol from Ref. 20.

If fracture occurs in a perfectly brittle manner, provided glasses behave purely elastic, the fracture toughness can be estimated from the following expression using the Irwin–Griffith similarity principle:²¹

$$K_c = (2E\gamma)^{1/2} \quad (10)$$

The results of the theoretical calculation are summarized in Table II. The calculated values are in relatively good agreement

Table II. Density, Elastic Moduli, Indentation Hardness, and Toughness of Ge–Se Glasses

Property	Property value for various Ge/Se [†]								
	0/100 [†]	5/95	10/90	15/85	20/80	25/75	30/70	40/60	100/0 [‡]
$\langle r \rangle$	2	2.1	2.2	2.3	2.4	2.5	2.6	2.7	4
ρ (g/cm ³)	4.28	4.31	4.34	4.36	4.37	4.36	4.32	4.36	5.15
E (GPa)	10.25	11.05	12.08	13.80	14.73	16.05	17.90	22.38	123.5
K (GPa)	9.60	10.01	10.43	11.22	11.47	12.21	12.64	16.43	70.95
ν	0.322	0.316	0.307	0.295	0.286	0.281	0.264	0.273	0.210
H (GPa)	0.39	0.57	0.77	1.05	1.38	1.72	2.02	2.35	7.37
H_V (GPa)	0.36	0.52	0.71	0.97	1.28	1.59	1.88	2.18	6.84
K_c (MPa·m ^{1/2})	0.16	0.09	0.12	0.22	0.28	0.22	0.20	0.16	0.59
$K_{c,\text{theo}}$ (MPa·m ^{1/2}) [§]	0.197	0.211	0.227	0.250	0.265	0.283	0.304	0.351	0.965
$K_{c,\text{theo}}$ (MPa·m ^{1/2}) [¶]	0.179	0.194	0.211	0.235	0.251	0.270	0.293	0.337	0.887

[†]Partially devitrified. [‡]Data obtained on polycrystalline germanium. [§]From Eqs. (8), (9), and (10) using the values for the bond energies from Ref. 15. [¶]From Eqs. (8), (9), and (10) using the values for the bond energies from Ref. 20.

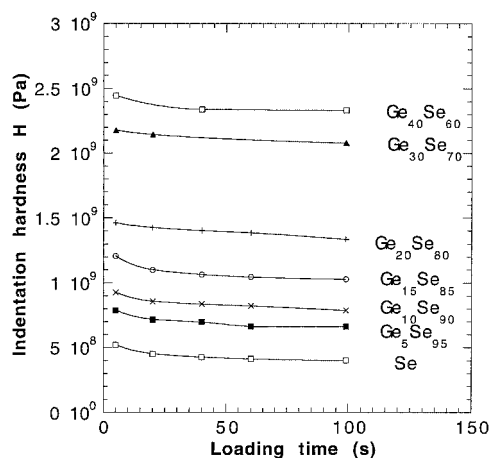


Fig. 8. Dependence of the indentation hardness on the loading time.

with the experimental results, especially for glasses with 15%–40% germanium. According to this theoretical analysis, increasing fracture toughness with increasing germanium contents between 5 and 20 at.% is thus supposed to result from increasing fraction of Ge–Se bonds and the corresponding increase of the network cross-linking. However, as the germanium content increases further from 20% to 30%, the glass density and the fracture toughness decrease. This tendency is attributed to the formation of Ge–Se–Ge bridges, which decrease the flexibility of the network, and, more particularly, of the selenium chains, and, thus, impede structural (global) and crack-tip (local) relaxation phenomena. The $\text{Ge}_{20}\text{Se}_{80}$ composition is associated to $\langle r \rangle = 2.4$ and corresponds to the rigidity percolation threshold.^{7,12} The increase of the wavenumber associated with a vibrational mode of the selenium chains at $\sim 260 \text{ cm}^{-1}$ ²² with increasing germanium content reflects this increase of the atomic network rigidity. For germanium contents $>25\%$, the fracture toughness, as predicted by the calculation, is slightly overestimated. A plausible explanation is that fracture follows low-energy paths, meeting either weak intermolecular van der Waals forces or Ge–Ge bonds. The two-dimensional layer structures consisting of a layer of germanium atoms bridging two selenium layers are fully evolved for the GeSe_2 glass, i.e., for $\langle r \rangle = 2.67$. These sandwich units are separated by weak van der Waals bonds resulting from the unpaired electrons of the selenium atoms of the outer layers. At lower $\langle r \rangle$ values, the presence of the selenium chains leads to a more-interwoven network structure, and, at higher $\langle r \rangle$ values, the transition to a three-dimensional network structure starts with the appearance of Ge–Ge bonds. In both cases, higher

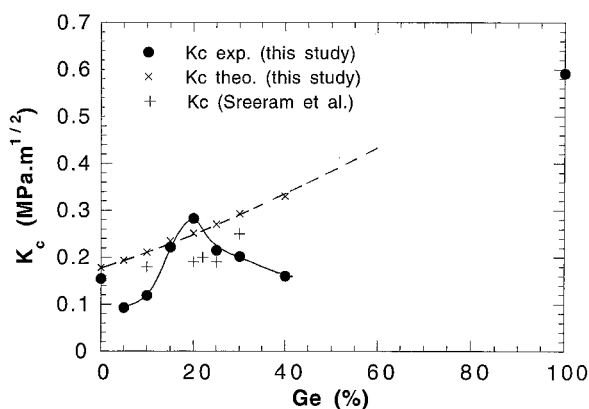


Fig. 9. Indentation toughness, K_c , of Ge–Se glasses as a function of the germanium content.

values of the fracture toughness are expected, consistent with our experimental results.

Similar to the hardness measurements, toughness measurements depend on the loading duration. Although the toughness ranking remains unchanged, K_c decreases with increasing loading time (Fig. 10).

(5) Scratch Resistance and Brittleness

Three compositions of Ge–Se glasses have been selected for this study. These compositions are the $\text{Ge}_{10}\text{Se}_{90}$, $\text{Ge}_{20}\text{Se}_{80}$, and $\text{Ge}_{30}\text{Se}_{70}$ grades and correspond to the low-toughness region in the chalcogen-rich range, to the maximum toughness, and to the low-toughness region on the germanium-rich range, respectively. These glasses behave in significantly different ways (Fig. 11). The $\text{Ge}_{10}\text{Se}_{90}$ grade is the softest of the three grades, and a relatively large groove results from the scratch experiment, with subsurface lateral cracks extending over relatively large distances, some of them intersecting the surface and leading to the formation of chips. On the contrary, cracks extend little from the groove in the $\text{Ge}_{20}\text{Se}_{80}$ grade, and chips are small and remain on the border of the microplastic groove. The scratch in the $\text{Ge}_{30}\text{Se}_{70}$ grade is particularly chaotic, with the formation of much debris and large chips because of relatively long radial and lateral cracks (the $\text{Ge}_{30}\text{Se}_{70}$ grade is harder but has a lower fracture toughness than the $\text{Ge}_{20}\text{Se}_{80}$ grade). To analyze the scratch resistance in a more quantitative manner, observations must be discussed as a result of the hardness, which controls the width and the depth of the track; the toughness, which controls the crack initiation and propagation; and of more-complex quantities combining hardness, fracture toughness, and Young's modulus, which are discussed in the next paragraph.

IV. Discussion: Surface Damage and Brittleness of Ge–Se Glasses

On the one hand, Ge–Se glasses can be considered as model brittle materials under normal service conditions because of their extremely low fracture toughness, especially for chalcogen-rich composition. On the other hand, these glasses are much softer than most other glasses, and, therefore, they accommodate surface stress concentration through a viscoplastic flow process. Thus, the brittleness of these glasses depends on the competition between the increasing stress intensity during the loading stage and the stress relaxation ability, which reflects through the hardness. The radial-median and the lateral crack lengths control the surface damage; the lateral cracks are

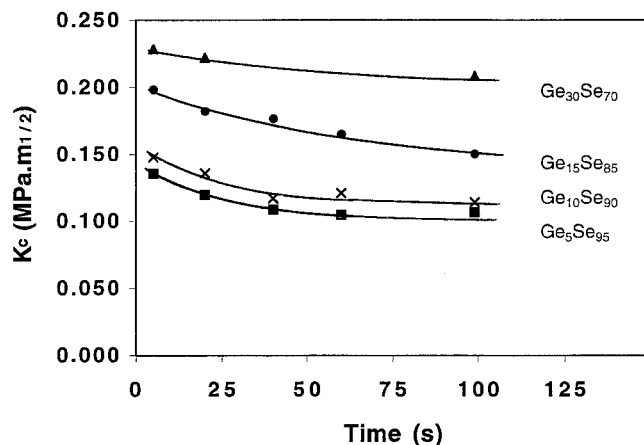


Fig. 10. Dependence of the indentation fracture toughness measurement on the loading time.

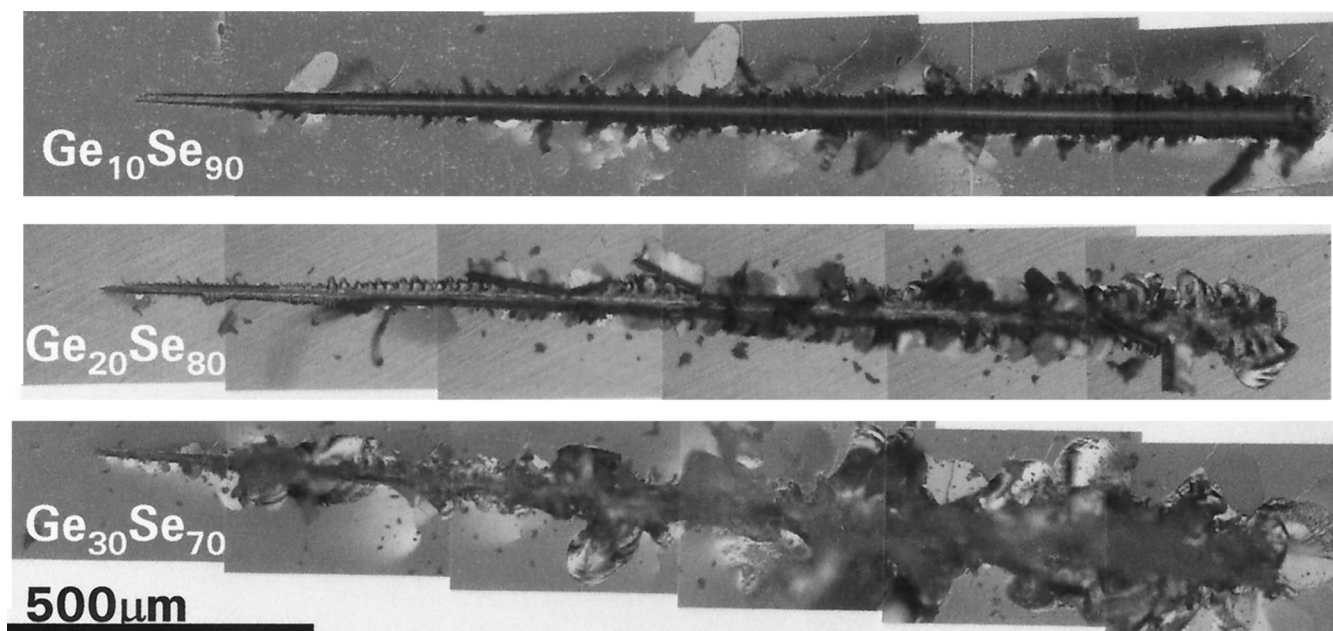


Fig. 11. Optical microscopy observations of the grooves left on the $\text{Ge}_{10}\text{Se}_{90}$, $\text{Ge}_{20}\text{Se}_{80}$, and $\text{Ge}_{30}\text{Se}_{70}$ glasses after scratch experiments conducted using a diamond cone indenter (apex angle = 90°) and with a loading rate of 0.01 N/s.

predominant on the chipping process. When $P > P^*$, the radial-median crack length can be derived from Eq. (6):

$$c \propto \left(\frac{E}{H}\right)^{1/3} \left(\frac{1}{K_c}\right)^{2/3} P^{2/3} \quad (11)$$

An expression for the lateral crack length has been proposed by Evans and Wilshaw:²³

$$c_l \propto \left(\frac{P}{K_c}\right)^{3/4} \quad (12)$$

All the governing material factors are summarized in Table III. In addition, the E/H ratio, which allows for an estimation of the magnitude of the residual elastic stress field in sharp-contact fracture,²⁴ as well as the brittleness parameter, $B = H/K_c$, previously introduced by Lawn and Marshall,¹³ are also indicated in Table III. Some striking results appear from Table III: (i) whereas the fracture toughness of selenium-rich glasses is remarkably low, for instance in comparison with a standard float glass ($K_c = 0.76 \text{ MPa}\cdot\text{m}^{1/2}$), the brittleness B is much lower than that of the float glass ($B = 8356 \text{ m}^{-1/2}$); and (ii) the large extension of the radial-median cracks results from a low fracture toughness and a high E/H ratio ($E/H = 9.7$ for a float glass) (see Eq. (12)). Another consequence of a high E/H ratio, especially for selenium-rich compositions, is the low magnitude of elastic recovery after unloading, so that the indentation inverse pyramids appear with flat surfaces and almost perfect square bases. In comparison with $\text{Ge}_{10}\text{Se}_{90}$ and $\text{Ge}_{30}\text{Se}_{70}$ grades, the $\text{Ge}_{20}\text{Se}_{80}$ grade is characterized by a higher fracture toughness. Although its hardness and Young's modulus are

intermediate, the radial-median and the lateral crack extensions are smaller for the $\text{Ge}_{20}\text{Se}_{80}$ grade. Therefore, as shown in Fig. 11, chipping appears at higher load, and fewer chips form on the surface of this glass in comparison with others during scratch experiments.

V. Conclusion

The usual trend in materials science is that the harder the material, the lower its fracture toughness. With this regard, Ge–Se glasses behave unusually, because they exhibit at the same time low hardness and low fracture toughness. Their hardness is between 0.39 and 2.35 GPa and increases continuously with germanium content. The fracture toughness follows a more-complicated compositional dependence with a maximum of $0.28 \text{ MPa}\cdot\text{m}^{1/2}$ at 20% germanium content and values as low as $0.1 \text{ MPa}\cdot\text{m}^{1/2}$ in the selenium rich side. Hardness and fracture toughness trends have been successfully discussed as a result of structural models accounting for the topological changes occurring in the Ge–Se glass network with increasing germanium contents. The scratch behavior is more difficult to address, because it combines hardness, toughness, and Young's modulus. Of particular interest is the fact that, because of a high E/H ratio, the studied chalcogenide glasses develop high residual stress fields under sharp mechanical contacts. Moreover, despite their low fracture toughness values and their strong propensity to behave in a very brittle manner, Ge–Se glasses exhibit surprisingly low values of the brittleness parameter (B). Although this result finds its source in the low hardness of Ge–Se glasses, it shows that the B parameter, which

Table III. Factors Controlling Indentation Fracture and the Scratch Behavior

Factor	Factor values for various Ge/Se								
	0/100	5/95	10/90	15/85	20/80	25/75	30/70	40/60	100/0 [†]
E/H	26.3	19.4	15.7	13.1	10.7	9.3	8.9	9.5	16.8
$K_c^4/H^3 \times 10^7$	97.3	4.04	4.39	21.0	24.4	4.20	2.02	0.50	3.07
$(K_c/H)^2 \times 10^8$	15.8	2.66	2.39	4.47	4.21	1.56	1.00	0.46	0.65
$(E/H)^{1/3} K_c^{-2/3} \times 10^4$	10.3	13.1	10.3	6.4	5.1	5.9	6.0	7.2	3.6
$B \text{ (m}^{-1/2}\text{)}$	2516	6129	6470	4730	4876	8000	10000	14688	12449

[†]Crystallized germanium.

was proposed earlier to evaluate brittleness of ceramics and glasses, is inadequate to compare glasses from different families having much different bond characters or refractoriness.

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